

Sunanda Tata

San Jose, CA | +1 (650) 764-9023 | sunandatata.careers@gmail.com | linkedin.com/in/tata-sunanda | github.com/sunandatata

Education

California State University, Fullerton
Master of Science in Computer Science
GPA: 3.7 / 4.0

Fullerton, CA
Aug 2024 – May 2026

Skills

Programming Languages: Python, Java, JavaScript, TypeScript, Go, SQL, HTML/CSS

AI & Machine Learning: Generative AI, Large Language Models (LLMs), Machine Learning, Natural Language Processing (NLP), LangChain, Retrieval-Augmented Generation (RAG), AI Agents, Prompt Engineering, Embeddings, Semantic Search, Vector Databases, Knowledge Graphs, RoBERTa, T5

Full-Stack & Backend: React.js, Node.js, FastAPI, Spring Boot, Spring WebFlux, REST APIs, Microservices, JSON, PostgreSQL, MySQL, MongoDB, Redis

Cloud & DevOps: AWS, Azure, Google Cloud Platform (GCP), Docker, Kubernetes, Apache Kafka, Git, GitHub Actions, Jenkins, Terraform, CI/CD, Grafana, Prometheus

Software Engineering: Data Structures, Algorithms, Object-Oriented Programming (OOP), Distributed Systems, Database Systems, Enterprise Applications, ERP, Debugging, Code Review, Unit Testing, Integration Testing, Test-Driven Development (TDD), SDLC, Agile, Scrum

Experience

Pacific Life
Software Developer Intern

Newport Beach, CA
Jun 2025 – Mar 2026

- Reduced manual geospatial review effort by **75%** by building a **Python underwriting intelligence platform** integrating ESRI REST APIs, LangChain, and LLM services for regional underwriting analysis.
- Reduced external geocoding API usage by **35%** by implementing a **Redis cache-first architecture**, lowering third-party request volume and API costs.
- Reduced underwriting policy lookup time from **15 minutes to 30 seconds per case** by building a **Retrieval-Augmented Generation (RAG)** pipeline using embeddings, vector search, and grounded responses.
- Reduced LLM inference costs by **28%** by routing **70% of routine requests to Phi-3 Mini** and reserving larger language models for complex policy-reasoning tasks.
- Reduced exception-resolution time from **8 minutes to 3 minutes per case** by building **REST APIs** and **React/TypeScript** workflows exposing policy context, model confidence, and remediation actions.
- Maintained **99.9% workflow availability** by deploying services on **Kubernetes** with retries, health checks, cache fallbacks, automated testing, CI/CD, and human-in-the-loop safeguards.
- Reduced changed-record data latency to **under 24 hours** through overnight incremental processing and monthly portfolio reconciliation across underwriting, risk, and reporting systems.

Hitachi Vantara
Software Engineer

Bengaluru, India
Aug 2022 – Aug 2024

- Scaled **Lumada Data Catalog** to process and expose **1M+ enterprise metadata records** by building **Java/Spring Boot REST APIs** across distributed databases and heterogeneous data sources.
- Achieved **99.9% service availability** by engineering fault-tolerant metadata ingestion and persistence with **Spring Integration, JPA, and Hibernate**, backed by TDD and automated recovery.
- Reduced metadata synchronization time by **45%** by building asynchronous migration and replication pipelines with **Spring Batch and Apache Kafka**, keeping catalog data consistent across distributed environments.
- Reduced metadata retrieval latency by **40%** by developing reactive **Spring WebFlux** services and **React.js** workflows for metadata search, discovery, and visualization.
- Reduced regression-testing cycles by **60%** by automating Lumada Data Catalog deployment and validation using **AWS, Docker, Kubernetes, Jenkins, integration testing, and CI/CD**.

Hitachi Vantara
Software Engineer Co-Op

Bengaluru, India
Jan 2022 – Aug 2022

- Improved query-processing efficiency by **15%** by contributing to backend development for the **Pentaho platform** using **Java and SQL**, while supporting system architecture and technical design.
- Delivered **8+ technical demonstrations** to stakeholders during Agile sprints, communicating platform capabilities, implementation progress, and engineering design decisions.
- Reduced environment setup and regression-testing cycles by **50%** by supporting containerized deployments with **Docker** and developing automated integration tests in **Python**.

Open Source Contributions

RAGFlow

github.com/infiniflow/ragflow

Open Source Contributor

- Improved **retrieval API correctness across 20+ regression scenarios** by fixing parameter-precedence handling for top-k retrieval, similarity thresholds, reranking, and saved search configurations.
- Restored **100% correct top-k ordering across 25+ reranking test cases** by fixing the **RAG retrieval pipeline** to sort reranked chunks before pagination, preventing high-relevance results from being excluded from LLM context.
- Improved **multilingual retrieval consistency across 3 languages** by extending Agent memory retrieval with cross-language support and validating dataset and memory retrieval behavior through **15+ regression tests**.

Projects

Ride Matching Platform

github.com/sunandatata/Ride-Matching-Platform

- Built an end-to-end ride-dispatch platform across **3 role-specific applications** by engineering **React/TypeScript** rider, driver, and admin clients with **Node.js/Express REST APIs** and real-time trip synchronization.
- Implemented a **7-stage real-time ride lifecycle** from booking and driver matching through pickup, trip completion, and analytics using **WebSockets, live ETA updates, route visualization, and synchronized state**.
- Designed **7 backend domain boundaries** across authentication, rides, drivers, matching, location, ETA, and notifications, enabling modular APIs and independently extensible mobility workflows.
- Strengthened end-to-end reliability with **11+ automated test scripts** covering ride requests, driver acceptance, pricing, synchronization, application behavior, and complete ride flows, with **Docker-based deployment**.

AI²-SED Defense Framework

- Developed an adversarial **machine learning** framework for phishing detection using **RoBERTa** and URL-based features, with an adaptive attack pipeline powered by **T5**.
- Exposed model inference through a **FastAPI REST API** and developed a **Streamlit** interface with structured JSON logging for experiment analysis and model evaluation.